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Introduction

1.1 Probability Spaces

Definition 1.1.1 ► Probability Spaces

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \rightarrow [0, 1]$$

Example 1.1.2 ► Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

Example 1.1.3 ► Measures on the real line.

Measures on $(\mathbf{R}, \mathcal{R})$ are defined by giving probability a **Stieltjes measure function** with the following properties:

- (i) F is nondecreasing.
- (ii) F is right continuous, that is, $\lim_{y \downarrow x} F(y) = F(x)$.

Theorem 1.1.4

Associated with each Stieltjes measure function F there is a unique measure μ on $(\mathbf{R}, \mathcal{R})$ with $\mu((a, b]) = F(b) - F(a)$

$$\mu((a, b]) = F(b) - F(a) \tag{1.1}$$

When $F(x) = x$ the resulting measure is called **Lebesgue measure**

Remark.

The proof is long and winding. We will show the main idea following.

Definition 1.1.5 ▶ Semialgebra

A collection $\mathcal{S}$ of sets is said to be a **semialgebra** if (i) it is closed under intersection, that is, $S, T \in \mathcal{S}$ implies $S \cap T \in \mathcal{S}$, and (ii) if $S \in \mathcal{S}$ then S^c is a finite disjoint union of sets in $\mathcal{S}$.

Example 1.1.6

$S_d =$ the empty set plus all sets of the form

$$(a_1, b_1] \times \cdots \times (a_d, b_d] \subset \mathbb{R}^d \quad \text{where } -\infty \leq a_i < b_i \leq \infty$$

Then S_d is a semialgebra.

Remark.

The above Stieljes measure function gives the values of μ on the semialgebra.

Definition 1.1.7 ▶ Algebra

A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** (or **field**) if $A, B \in \mathcal{A}$ implies A^c and $A \cup B$ are in $\mathcal{A}$.

Remark.

Since $A \cap B = (A^c \cup B^c)^c$, it follows that $A \cap B \in \mathcal{A}$. Obviously a σ -algebra is an algebra.

Lemma 1.1.8

If $\mathcal{S}$ is a semialgebra, then $\bar{\mathcal{S}} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$ is an algebra, called the **algebra generated by $\mathcal{S}$** .

Remark.

Given a set function μ on $\mathcal{S}$, we can extend it to $\bar{\mathcal{S}}$ by

$$\mu\left(+_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i)$$

Proof. Suppose $A = +_i S_i$ and $B = +_j T_j$, where $+$ denotes disjoint union and we assume the index sets are finite. Then $A \cup B = +_{i,j} S_i \cap T_j$, where $S_j \cap T_j \in \mathcal{S}$

by definition. Thus $\bar{\mathcal{S}}$ is closed under intersection. As for complements, if $A = +_i \mathcal{S}_i$, then $A^c = \cap_i \mathcal{S}_i^c$, where $\mathcal{S}_i^c$ is a finite disjoint union of sets in $\mathcal{S}$, thus is in $\bar{\mathcal{S}}$. Since $\bar{\mathcal{S}}$ is closed under intersection, then $A^c \in \bar{\mathcal{S}}$ by induction. $\square$

Definition 1.1.9 ► measure on an algebra

By a **measure on an algebra** $\mathcal{A}$, we mean a set function μ with

- (i) $\mu(A) \geq \mu(\emptyset) = 0$ for all $A \in \mathcal{A}$, and
- (ii) if $A_i \in \mathcal{A}$ are **disjoint** and **their union is in** $\mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 1.1.10 ► σ -finite

μ is said to be **σ -finite** if there is a sequence of sets $A_n \in \mathcal{A}$ so that $\mu(A_n) < \infty$ and $\cup_n A_n = \Omega$.

Remark.

Letting $A'_1 = A_1$ and for $n \geq 2$,

$$A'_n = \bigcup_{m=1}^n A_m \quad \text{or} \quad A'_n = A_n \cap \left(\bigcap_{m=1}^{n-1} A_m^c \right) \in \mathcal{A}$$

we can without loss of generality assume that $A_n \uparrow \Omega$ or the A_n are disjoint.

Theorem 1.1.11

Let $\mathcal{S}$ be a semialgebra and let μ defined on $\mathcal{S}$ have $\mu(\emptyset) = 0$. Suppose (i) if $S \in \mathcal{S}$ is a finite disjoint union of sets $S_i \in \mathcal{S}$ then $\mu(S) = \sum_i \mu(S_i)$, and (ii) if $S_i, S \in \mathcal{S}$ with $S = +_{i \geq 1} S_i$ then $\mu(S) \leq \sum_{i \geq 1} \mu(S_i)$. Then μ has a unique extension $\bar{\mu}$ that is a measure on $\bar{\mathcal{S}}$ the algebra generated by $\mathcal{S}$. If $\bar{\mu}$ is sigma-finite then there is a unique extension ν that is a measure on $\sigma(\mathcal{S})$.

Lemma 1.1.12

Suppose only that (i) holds.

1. If $A, B_i \in \bar{\mathcal{S}}$ with $A = +_{i=1}^n B_i$, then $\bar{\mu}(A) = \sum_i \bar{\mu}(B_i)$.

2. If $A, B_i \in \bar{\mathcal{S}}$ with $A \subseteq \bigcup_{i=1}^n B_i$ then $\bar{\mu}(A) \leq \sum_i \bar{\mu}(B_i)$

Measures on $\mathbb{R}^d$

1.1.1 Exercice

Exercise 1.1.13

A set $A \subset \{1, 2, \dots\}$ is said to have *asymptotic density* θ if

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n = \theta$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a σ -algebra? an algebra?

Proof. Suppose that $A \in \mathcal{A}$ has asymptotic density θ .

$$|A \cap \{1, 2, \dots, n\}|/n + |A^c \cap \{1, 2, \dots, n\}|/n = 1$$

Take the limit of $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n + \lim_{n \rightarrow \infty} |A^c \cap \{1, \dots, n\}|/n = 1$$

which implies that A^c has the asymptotic density $1 - \theta$.

Let $A = \{2k\}_{k=1}^{\infty}$. Let B contained all the even numbers among $[2^{2m}, 2^{2m+1} - 1]$, and contained all the odd numbers among $[2^{2m+1}, 2^{2m+2} - 1]$. Since there are as many even numbers as odd numbers among $[2^k, 2^k - 1]$. It is not hard to see B has asymptotic density $\frac{1}{2}$. Then

$$\begin{aligned} \lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+1}\}|/2^{2m+1} &= \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2k+1} - 2^{2k})/2^{2m+1} \\ &= \lim_{m \rightarrow \infty} \sum_{k=0}^m 4^{k-m-1} \\ &= \lim_{m \rightarrow \infty} 4^{-m-1} \frac{1 - 4^{m+1}}{1 - 4} = \frac{1}{3} \end{aligned}$$

$$\lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+2}\}| / 2^{2m+2} = \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2m+1} - 2^{2m}) / 2^{2m+2} = \frac{1}{6}$$

which implies that $A \cap B$ is not in $\mathcal{A}$. But $\mathcal{A}$ is closed under complements, $\mathcal{A}$ is not closed under finite union. Thus $\mathcal{A}$ is not an algebra, then is neither a σ -algebra. $\square$

1.2 Distributions

Definition 1.2.1

A real-valued function X defined on Ω is said to be a **random variable** if for every Borel set $B \subseteq \mathbb{R}$ we have $X^{-1}(B) \in \mathcal{F}$. That is a real valued measurable function on $\mathcal{F}$. We write $X \in \mathcal{F}$.

Example 1.2.2

1. If Ω is a discrete probability space, then any function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.
2. For any $A \in \mathcal{F}$, the **indicator function** of A is:

$$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

Definition 1.2.3 ► distribution

- If X is a random variable, then X induces a probability measure on $\mathbb{R}$ called **distribution** by setting $\mu(A) = P(X \in A)$ for Borel sets A .
- The distribution of a random variable X is usually described by giving its **distribution function**,

$$F(x) = P(X \leq x)$$

Remark.

The right-hand side can be written as $P(X^{-1}(A))$. In words, we pull $A \in \mathbb{R}$ back to $X^{-1}(A) \in \mathcal{F}$ and then take P of that set.

Proof. If the A_i are disjoint,

$$\begin{aligned}\mu\left(\bigcup_i A_i\right) &= P(X \in A) \\ &= P\left(\bigcup_i \{X \in A_i\}\right) \\ &= \sum_i P(X \in A_i) = \sum_i \mu(A_i)\end{aligned}$$

□

Theorem 1.2.4

Any distribution function F has the following properties:

- (i) F is **nondecreasing**.
- (ii) $\lim_{x \rightarrow \infty} F(x) = 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$.
- (iii) F is **right continuous**, that is, $\lim_{y \downarrow x} F(y) = F(x)$.
- (iv) If $F(x-) = \lim_{y \uparrow x} F(y)$ then $F(x-) = P(X < x)$.
- (v) $P(X = x) = F(x) - F(x-)$.

Theorem 1.2.5

If F satisfies (i), (ii), and (iii) in the above theorem, then it is the distribution function of some random variable.

Remark.

Even though F may not be 1-1 and onto, we will call X the inverse of F and denote by F^{-1} .

Proof. Let P be Lebesgue measure, Note that

$$F(x) = P(\omega : \omega \leq F(x))$$

And if X is a random variable which we need, then

$$F(x) = P(\omega : X(\omega) \leq x)$$

Thus we only need to show that

$$\{\omega : X(\omega) \leq x\} = \{\omega : \omega \leq F(x)\}$$

holds for some X . If F is a bijection, we may define

$$X(\omega) = F^{-1}(\omega)$$

It motivates us to define

$$X(\omega) = \sup \{y : F(y) < \omega\}$$

in a general case. To check that it is just what we need, we observe that if $\omega \leq F(x)$, and if $x \leq X(\omega)$, then $F(x) < \omega$, which is a contradiction. Thus $x > X(\omega)$. On the other hand if $\omega > F(x)$, then since F is right continuous, there is an $\varepsilon > 0$ so that $F(x + \varepsilon) < \omega$, and $X(\omega) \geq x + \varepsilon > x$. $\square$

Definition 1.2.6 ► Equal in Distribution

- If X and Y induce the same distribution μ on $(\mathbb{R}, \mathcal{R})$, we say X and Y are **equal in distribution**, written by

$$X =_d Y$$

- When the distribution function $F(x) = P(X \leq x)$ has the form

$$F(x) = \int_{-\infty}^x f(y) \, dy$$

We say that X has **density function** f . We write $f_X(x)$.

Remark.

- X and Y are equal in distribution, iff X and Y have the same distribution function, that is, $P(X \leq x) = P(Y \leq x)$ for all x .
- In remembering formulas, it is often useful to think of $f(x)$ as being $P(X = x)$ although

$$P(X = x) = F(x) - F(x-) = \lim_{\varepsilon \rightarrow 0} \int_{x-\varepsilon}^{x+\varepsilon} f(y) \, dy = 0$$

Example 1.2.7 ▶ Uniform distribution on (0,1)

$f(x) = 1$ for $x \in (0, 1)$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Example 1.2.8 ▶ Exponential distribution with rate λ

$f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Example 1.2.9 ▶ Standard normal distribution

$$f(x) = (2\pi)^{-1/2} \exp(-x^2/2)$$

Remark.

In this case, there is no closed-form expression for $F(x)$, but we have the following bounds that are useful for large x :

Theorem 1.2.10 ▶ For $x > 0$

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2)$$

1.3 Random Variables

Definition 1.3.1

A function $X : \Omega \rightarrow S$ is said to be a **measurable map** from $(\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ if

$$X^{-1}(B) \equiv \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}$$

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$ and $d > 1$, then X is called a **random vector**. If $d = 1$, X is called a **random variable**, ro r.v. for short.

Theorem 1.3.2

If $\{\omega : X(\omega) \in A\} \in \mathcal{F}$ for all $A \in \mathcal{A}$ and $\mathcal{A}$ **generates** $\mathcal{S}$, then X is measurable.

Definition 1.3.3

Suppose that X is a measurable map. Then we define

$$\sigma(X) := \{\{X \in B\} : B \in \mathcal{S}\}$$

as the **σ -field generated by X** .

Remark.

where we write $\{X \in B\}$ as shorthand for $\{\omega : X(\omega) \in B\}$.

Example 1.3.4

If $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{R})$, then possible choices of $\mathcal{A}$ in 1.3.3 are $\{(-\infty, x] : x \in \mathbb{R}\}$ or $\{(-\infty, x) : x \in \mathbb{Q}\}$.

Example 1.3.5

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$, a useful choice of $\mathcal{A}$ is

$$\{(a_1, b_1) \times \cdots \times (a_d, b_d) : -\infty < a_i < b_i < \infty\}$$

Theorem 1.3.6

If $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ and $f : (S, \mathcal{S}) \rightarrow (T, \mathcal{T})$ are measurable maps, then $f(X)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(T, \mathcal{T})$.

Theorem 1.3.7

If $X_1, \dots, X_n$ are random variables and $f : (\mathbb{R}^n, \mathcal{R}^n) \rightarrow (\mathbb{R}, \mathcal{R})$ is measurable, then $f(X_1, \dots, X_n)$ is a random variable.

Theorem 1.3.8

If $X_1, \dots, X_n$ are random variables then $X_1 + \cdots + X_n$ is a random variable.

Theorem 1.3.9

f $X_1, X_2, \dots$ are random variables then so are

$$\inf_n X_n \quad \sup_n X_n \quad \limsup_n X_n \quad \liminf_n X_n$$

Exercise**Exercise 1.3.10**

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then $X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$ generates $\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}$

Proof. First, we show that

$$\sigma(X^{-1}(\mathcal{A})) \subseteq X^{-1}(\mathcal{S})$$

For which, we only need to show that

1. $X^{-1}(\mathcal{S})$ is a σ -algebra;
 2. $X^{-1}(\mathcal{A}) \subseteq X^{-1}(\mathcal{S})$.
2. is obvious from $\mathcal{A} \subseteq \mathcal{S}$. Since $\mathcal{S}$ is a σ -algebra, and the complementation, union are kept under preimage, then $X^{-1}(\mathcal{S})$ is a σ -algebra.

Next, we show that

$$X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$$

We define

$$\mathcal{C} := \{B \in \mathcal{S} : X^{-1}(B) \in \sigma(X^{-1}(\mathcal{A}))\}$$

It is not hard to see that $\mathcal{C}$ is a σ -algebra. Once we show that $\mathcal{A} \subseteq \mathcal{C}$, we have $\mathcal{S} \subseteq \mathcal{C}$, then it follows that $X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$. For that, we take $A \in \mathcal{A}$, since $X^{-1}(A) \in X^{-1}(\mathcal{A}) \subseteq \sigma(X^{-1}(\mathcal{A}))$, $A \in \mathcal{C}$. Thus we completes the proof. $\square$

1.4 Expected Value

Definition 1.4.1 ► Expected Value

- If $X \geq 0$ is a random variable on $(\Omega, \mathcal{F}, P)$, then we define its **expected value** to be

$$EX = \int X \, dP$$

We declare that EX **exists** and set

$$EX = EX^+ - EX^-$$

whenever the subtraction makes sense.

Remark.

EX is often called the **mean** of X and denoted by μ .

Theorem 1.4.2

Suppose $X, Y \geq 0$ or $E|X|, E|Y| < \infty$.

- $E(X + Y) = EX + EY$.
- $E(aX + b) = aE(X) + b$ for any real numbers a, b .
- If $X \geq Y$ then $EX \geq EY$.

1.4.1 Inequalities

Theorem 1.4.3 ► Jensen

Suppose φ is **convex**. Then

$$E(\varphi(X)) \geq \varphi(EX)$$

provided both expectations exist, that is, $E|X|$ and $E|\varphi(X)| < \infty$

Theorem 1.4.4 ► Holder

If $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$, then

$$E|XY| \leq \|X\|_p \|Y\|_q$$

Here $\|X\|_r = (E|X|^r)^{\frac{1}{r}}$ for $r \in [1, \infty)$; $\|X\|_\infty = \inf\{M : P(|X| > M) = 0\}$

1.4.2 Integration to the Limit

Theorem 1.4.5 ▶ Fatou's lemma

If $X_n \geq 0$ then

$$\liminf_{n \rightarrow \infty} EX_n \geq E\left(\liminf_{n \rightarrow \infty} X_n\right)$$

Theorem 1.4.6 ▶ Monotone convergence theorem

If $0 \leq X_n \uparrow X$ then $EX_n \uparrow EX$.

Theorem 1.4.7 ▶ Dominated convergence theorem

If $X_n \rightarrow X$ a.s., $|X_n| \leq Y$ for all n , and $EY < \infty$, then $EX_n \rightarrow EX$.

1.4.3 Computing Expected Values

Theorem 1.4.8 ▶ Change of variables formula

Let X be a random element of $(S, \mathcal{S})$ with distribution μ , that is, $\mu(A) = P(X \in A)$. If f is a measurable function from $(S, \mathcal{S})$ to $(\mathbb{R}, \mathcal{R})$ so that $f \geq 0$ or $E|f(X)| < \infty$, then

$$Ef(X) = \int_S f \, d\mu =: \int_S f(y) \mu(dy)$$

Definition 1.4.9

- If k is a positive integer, then EX^k is called the ***kth moment*** of X . The first moment EX is usually called the ***mean*** and denoted by μ .
- If $EX^2 < \infty$, then the ***variance*** of X is defined to be

$$\begin{aligned} \text{var}(X) &= E(X - \mu)^2 \\ &= EX^2 - \mu^2 \end{aligned}$$

Proposition 1.4.10

- $\text{var}(X) \leq EX^2$
- $\text{var}(aX + b) = a^2 \text{var}(X)$

Example 1.4.11

If X has the *exponential distribution* with rate 1, then

$$EX^k = \int_0^{\infty} x^k e^{-x} dx = k!$$

Proof.

$$\begin{aligned} EX^k &= \int_S y^k d\mu = \int_S y^k dF = \int_{\mathbb{R}} x^k f(x) dx = \int_0^{\infty} x^k e^{-x} dx \\ \int_0^{\infty} x^k e^{-x} dx &= - \int_0^{\infty} x^k de^{-x} = -[e^{-x} x^k]_0^{\infty} + k \int_0^{\infty} x^{k-1} e^{-x} dx \\ &= k \int_0^{\infty} x^{k-1} e^{-x} dx \end{aligned}$$

Then it follows from induction. □